

SEMESTER END EXAMINATIONS – JUNE / JULY 2026

Program : B.E. – Information Science and Engineering
Course Name : Machine Learning
Course Code : 23IS62

Semester : VI
Max. Marks : 100
Duration : 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Describe the complete Machine Learning process with a neat diagram. CO1 (06)
 - A binary classifier produced the following confusion matrix: CO1 (08)

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	35

- Calculate Accuracy, Precision, Recall, and F1-score.
 - Interpret the performance of the model.
 - Comment on whether the model is suitable for critical applications like medical diagnosis.
- Explain how bias and variance change with model complexity using a suitable example. CO1 (06)
 - Compare supervised, unsupervised, and reinforcement learning with one simple example CO1 (07)
 - A machine learning model is developed using a dataset and produces the following results: CO1 (07)
 - Training Accuracy = 98%
 - Validation Accuracy = 72%

Based on the above information, answer the following:

- Determine whether the model is overfitting or underfitting.
 - Suggest at least two methods to improve the model performance.
 - Explain the role of training, validation, and testing datasets in identifying such issues.
- Discuss the challenges posed by unbalanced datasets and methods to handle them. CO1 (06)

UNIT - II

- Differentiate data matrix and dissimilarity matrix with an example. CO2 (06)
 - Discuss the steps of the algorithm and how S and G boundaries are updated for Positive examples and Negative examples. CO2 (08)
 - Consider the following dataset CO2 (06)

Student	Study	Attendance	Assignment	Result
S1	High	Good	Yes	Pass
S2	Low	Poor	No	Fail
S3	Medium	Good	Yes	Pass
S4	High	Poor	Yes	Pass
S5	Low	Good	No	Fail
S6	Medium	Poor	Yes	Pass

Frame the concept learning task and identify, Instance space, Hypothesis space and Target concept.

4. a) Consider the dataset

CO2 (06)

Object	Height (cm)	Weight (kg)	Color	Shape	Class
A	160	55	Red	Round	Yes
B	170	65	Blue	Round	No
C	165	60	Red	Square	Yes

Convert categorical attributes into suitable form, normalize numeric attributes, Compute pairwise dissimilarity matrix using mixed-attribute distance and draw dissimilarity matrix.

- b) Consider the following dataset.

CO2 (06)

Color	Size	Shape	Texture	Target
Red	Small	Round	Smooth	Yes
Red	Large	Round	Smooth	Yes
Red	Small	Square	Smooth	Yes
Blue	Small	Round	Smooth	No
Red	Small	Round	Rough	Yes

Apply the Find-S algorithm step-by-step. Show hypothesis after each positive example, identify attributes that become generalized (?) and write the final hypothesis.

- c) Explain the working of LDA with suitable mathematical expressions.

CO2 (08)

UNIT - III

5. a) Compare simple linear regression and multiple linear regression with suitable examples. Using the following housing dataset, formulate a multiple linear regression equation to predict house price based on Size, Number of Bedrooms, and Age of the house.

CO3 (08)

Size (sq.ft)	Bedrooms	Age (years)	House Price (₹ Lakhs)
1200	2	10	50
1500	3	8	65
1800	3	6	80
2000	4	5	90

- b) For the Play-Tennis dataset below, build the complete ID3 Decision Tree. Show the Entropy and Information Gain computation for each attribute at each level.

CO3 (12)

Day	Outlook	Temp	Humidity	Wind	Play?
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes

6. a) A medical dataset is given below for predicting whether a patient has Flu:

CO3 (10)

Fever	Cough	Flu
Yes	Yes	Yes
Yes	No	Yes
No	Yes	No
Yes	Yes	Yes
No	No	No

Using Naïve Bayes Classifier, predict whether a patient with:
Fever = Yes, Cough = No has Flu.

Calculate:

- i. Prior probabilities
 - ii. Likelihood probabilities
 - iii. Posterior probability
 - iv. Final predicted class
- b) A bank fraud detection dataset contains transaction points: CO3 (10)
 Fraudulent:
 (2,5), (3,6), (4,5)
 Non-Fraudulent:
 (8,9), (9,8), (10,10)
 i. Build linear SVM classification boundary.
 ii. Identify support vectors.
 iii. If fraudulent patterns overlap, explain how non-linear SVM handles this case.

UNIT- IV

7. a) Apply the K-Means Clustering algorithm to group the following set of 2D points into K=2 clusters. CO4 (07)
 Data Points: P1(2,3), P2(5,6), P3(8,7), P4(1,4)
 Initial Centroids: C1 = (2,3) and C2 = (5,6)
 Show the cluster assignments for each point after one full iteration using Euclidean distance.
- b) Explain DBSCAN and how it defines Core points, Border points, and Noise. CO4 (07)
- c) Describe the Elbow Method for choosing the right value of 'k' in clustering. CO4 (06)
8. a) Describe the working of DIANA (Divisive) Hierarchical Clustering techniques with an example. CO4 (07)
- b) Given the following proximity (distance) matrix for three data points A, B, and C, perform hierarchical clustering using the Single-linkage technique. CO4 (07)
 Proximity Matrix:
- | | A | B | C |
|---|------|------|------|
| A | 0 | 0.15 | 0.25 |
| B | 0.15 | 0 | 0.20 |
| C | 0.25 | 0.20 | 0 |
- c) Explain the Expectation Maximization (EM) algorithm in the context of unsupervised learning. CO4 (06)

UNIT - V

9. a) Explain Principal Component Analysis (PCA) and its importance in unsupervised dimensionality reduction. CO5 (07)
- b) Describe Ensemble Methods. Explain how Bagging improves performance. CO5 (07)
- c) Describe three methods of Feature Subset Selection for model selection and evaluation. CO5 (06)
10. a) Compare Boosting and Bagging approaches. CO5 (07)
- b) Discuss Cross-Validation for model evaluation. CO5 (07)
- c) Explain the concept of Model Selection Using Statistical Tests of Significance. CO5 (06)
